

BAGAS AGNANDRI WIJAYA

Railway Infrastructure Maintenance Professional

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PROFESSIONAL SUMMARY

Railway Infrastructure Maintenance Professional with **6+ years of experience** in High-Speed Railway operations, specializing in Bridge, Tunnel & Subgrade Maintenance, Asset Management, Infrastructure Inspection, and Operation & Maintenance Readiness.

Currently serving as **Bridge, Tunnel & Subgrade Maintenance Deputy Foreman** at PT Kereta Cepat Indonesia China (KCIC), leading maintenance activities across a **61 km High-Speed Railway corridor** supporting Indonesia's first High-Speed Railway.

Developed **75+ technical documents**, contributed to **Minister of Transportation Regulation No. 7 of 2022**, and supported infrastructure reliability through maintenance planning, regulatory compliance, technical documentation, and continuous improvement.

Strong leadership, analytical, and stakeholder coordination skills with a commitment to safe, reliable, and efficient railway operations.

KEY ACHIEVEMENTS

- **6+ years** of professional experience in High-Speed Railway infrastructure maintenance and operations.
- Currently serving as **Bridge, Tunnel & Subgrade Maintenance Deputy Foreman** at PT Kereta Cepat Indonesia China (KCIC).
- Responsible for maintenance activities across a **61 km High-Speed Railway corridor**, including **27 bridges, 8 tunnels, 33 subgrade assets, 56 sound barriers, 12 height restriction portals, 7 wind monitoring stations, 4 Foreign Object Detection (FOD) systems, and 3 earthquake monitoring stations**.
- Supervised multidisciplinary maintenance teams and contractors to ensure safe, reliable, and efficient maintenance execution.
- Developed **75+ technical documents**, including Standard Operating Procedures (SOPs), Maintenance Procedures, Inspection Procedures, Work Instructions, and Preventive Maintenance Programs.
- Contributed to the development of **Minister of Transportation Regulation No. 7 of 2022** concerning High-Speed Railway Infrastructure.
- Supported **Operation & Maintenance Readiness**, Static Testing, Dynamic Testing, Defect Notification Period (DNP), and infrastructure handover for Indonesia's first High-Speed Railway.
- Experienced in Railway Safety Management System (**SMKP DJKA**), technical audits, maintenance governance, and regulatory compliance.
- Certified Railway Infrastructure Maintenance and Inspection Personnel for High-Speed Railway Bridge and Tunnel assets.

TECHNICAL EXPERTISE

Railway Infrastructure

- High-Speed Railway Infrastructure
- Bridge Engineering
- Tunnel Engineering
- Subgrade Engineering
- Infrastructure Inspection
- Permanent Way

Maintenance & Asset Management

- Preventive Maintenance
- Corrective Maintenance
- Maintenance Planning
- Maintenance Strategy
- Asset Lifecycle Management
- Asset Integrity
- Root Cause Analysis

Railway Operations

- Operation & Maintenance Readiness
- Railway Safety
- Maintenance Governance
- Regulatory Compliance
- SMKP DJKA

Leadership

- Team Leadership
- Contractor Management
- Cross-functional Coordination
- Stakeholder Management

PROFESSIONAL EXPERIENCE

PT KERETA CEPAT INDONESIA CHINA

BRIDGE, TUNNEL & SUBGRADE MAINTENANCE DEPUTY FOREMAN

September 2025 – Present | Jakarta – Bandung High-Speed Railway

Lead multidisciplinary maintenance activities to ensure the reliability, safety, and availability of critical railway infrastructure supporting Indonesia's first High-Speed Railway. Responsible for maintenance planning, execution, inspection, technical evaluation, contractor coordination, and regulatory compliance across a **61 km operational corridor**.

Key Responsibilities

- Lead and supervise preventive and corrective maintenance activities for critical infrastructure assets, including **27 bridges, 8 tunnels, 33 subgrade assets, 56 sound barriers, 12 height restriction portals, 7 wind monitoring stations, 4 Foreign Object Detection (FOD) systems, and 3 earthquake monitoring stations**.
- Coordinate multidisciplinary maintenance teams, contractors, and the **Operation Control Center (OCC)** to ensure maintenance activities are completed safely, efficiently, and with minimal impact on railway operations.
- Plan maintenance schedules, allocate manpower, prepare equipment and material requirements, and monitor work execution to achieve maintenance objectives.
- Evaluate inspection findings, identify infrastructure defects, determine maintenance priorities, and recommend corrective and preventive actions.
- Ensure maintenance activities comply with railway regulations, maintenance standards, occupational safety requirements, and the Railway Safety Management System (**SMKP DJKA**).
- Prepare maintenance reports, technical analyses, performance evaluations, and engineering recommendations to support management decision-making.
- Monitor contractor performance and work quality to ensure compliance with technical specifications, safety requirements, and project schedules.
- Support continuous improvement initiatives by identifying maintenance risks, improving work processes, and enhancing infrastructure reliability.

Key Achievements

- Successfully coordinated maintenance activities across multiple infrastructure disciplines, improving work execution efficiency and operational readiness.
- Strengthened coordination between maintenance teams, OCC, contractors, and internal stakeholders to improve maintenance planning and execution.
- Supported SMKP DJKA implementation and audit readiness through comprehensive maintenance documentation and regulatory compliance.
- Improved maintenance governance by standardizing work practices, technical documentation, and maintenance procedures.
- Led technical evaluations based on inspection findings, contributing to improved infrastructure reliability and long-term asset performance.

PT KERETA CEPAT INDONESIA CHINA

OFFICER – OPERATION & MAINTENANCE READINESS (Permanent Way & Building)

November 2019 – September 2025

Played a key role in preparing **Operation & Maintenance Readiness (O&M Readiness)** for Indonesia's first High-Speed Railway by developing maintenance systems, technical documentation, inspection procedures, maintenance standards, and operational readiness programs prior to commercial operation.

Key Responsibilities

- Contributed to the development of **Minister of Transportation Regulation No. 7 of 2022** by preparing technical concepts, comparative studies, regulatory analyses, and supporting documentation for High-Speed Railway infrastructure.
- Developed **75+ technical documents**, including Standard Operating Procedures (SOPs), Maintenance Procedures, Inspection Procedures, Work Instructions, Preventive Maintenance Programs, and Technical Guidelines.
- Supported **Static Testing, Dynamic Testing**, and operational readiness activities by preparing technical documentation and coordinating with internal departments and government authorities.
- Coordinated infrastructure inspections and monitored corrective actions during the **Defect Notification Period (DNP)** prior to infrastructure handover.
- Prepared maintenance resource planning, including tools, equipment, spare parts, and material requirements for operational readiness.

- Developed technical training materials and asset databases to support maintenance personnel competency development.
- Assisted in emergency preparedness verification, maintenance system implementation, and operational documentation.
- Participated in bridge maintenance supervision, bridge painting activities, and maintenance procedure development during assignment at **PT KAI Resort JRK Karawang**.

Key Achievements

- Contributed to the successful Operation & Maintenance Readiness of Indonesia's first High-Speed Railway.
- Developed more than **75 technical documents** supporting maintenance standardization and operational readiness.
- Contributed to the preparation of **Minister of Transportation Regulation No. 7 of 2022** concerning High-Speed Railway Infrastructure.
- Supported successful execution of Static Testing, Dynamic Testing, infrastructure inspections, and Defect Notification Period (DNP) activities.
- Established standardized maintenance documentation, inspection procedures, and preventive maintenance programs to support long-term infrastructure reliability.

DIRECTORATE GENERAL OF RAILWAYS (FIELD EXPERIENCE / PROJECT ASSIGNMENT) **RAILWAY TRACK CONSTRUCTION SUPERVISION & INFRASTRUCTURE TESTING** 2017 – 2019

Participated in railway infrastructure construction supervision and testing projects across multiple regions in Indonesia as part of academic field assignments and railway engineering development.

Key Responsibilities

- Supported railway track construction supervision within the Bandung Railway Engineering Office.
- Participated in Double Track railway infrastructure testing in Medan.
- Assisted in supervision of the Kebumen Double Track Project and 100-meter railway bridge construction in Central Java.
- Conducted field inspections, engineering documentation, and project administration activities in accordance with railway engineering standards.

MAJOR PROJECTS

Jakarta – Bandung High-Speed Railway (HSR) **Indonesia's First High-Speed Railway**

Project Owner: PT Kereta Cepat Indonesia China (KCIC)

Project Length: 61 km

Role: Operation & Maintenance Readiness Officer / Bridge, Tunnel & Subgrade Maintenance Deputy Foreman

Selected Contributions

- Contributed to the successful **Operation & Maintenance Readiness (O&M Readiness)** for Indonesia's first High-Speed Railway prior to commercial operation.
- Developed **75+ technical documents**, including Standard Operating Procedures (SOPs), Maintenance Procedures, Inspection Procedures, Work Instructions, Preventive Maintenance Programs, and Technical Guidelines.
- Supported **Static Testing, Dynamic Testing**, and infrastructure operational readiness by coordinating technical documentation and maintenance requirements.
- Coordinated infrastructure inspections and monitored corrective actions throughout the **Defect Notification Period (DNP)** before infrastructure handover.
- Currently responsible for supervising maintenance activities across a **61 km operational corridor**, ensuring infrastructure reliability, operational safety, and regulatory compliance.
- Coordinate multidisciplinary maintenance teams, contractors, and the Operation Control Center (OCC) to support safe and efficient railway operations.

Minister of Transportation Regulation No. 7 of 2022

High-Speed Railway Infrastructure Regulation

Role: Technical Contributor – High-Speed Railway Infrastructure

Selected Contributions

- Contributed to the development of technical concepts supporting the preparation of **Minister of Transportation Regulation No. 7 of 2022** concerning High-Speed Railway Infrastructure.
- Conducted comparative studies between conventional railway and high-speed railway infrastructure standards.
- Performed regulatory gap analysis and prepared supporting technical documentation for bridge, tunnel, and permanent way infrastructure.
- Collaborated with internal stakeholders and government representatives during the regulatory development process.

Railway Infrastructure Maintenance Improvement

Role: Bridge, Tunnel & Subgrade Maintenance Deputy Foreman

Selected Contributions

- Lead preventive and corrective maintenance programs for critical railway infrastructure assets supporting daily high-speed railway operations.
- Monitor infrastructure condition through inspections and technical evaluations to determine maintenance priorities and long-term asset reliability improvements.
- Support implementation of the Railway Safety Management System (**SMKP DJKA**) through maintenance documentation, audit preparation, and regulatory compliance.
- Drive continuous improvement initiatives by enhancing maintenance procedures, technical documentation, and work execution processes.

PROFESSIONAL CERTIFICATIONS

Directorate General of Railways (DJKA) – Ministry of Transportation, Republic of Indonesia

- **High-Speed Railway Infrastructure Maintenance Personnel – Bridge** Competency Certificate No. PRP.260897.73038
- **High-Speed Railway Infrastructure Maintenance Personnel – Tunnel** Competency Certificate No. PRP.260897.73039
- **High-Speed Railway Infrastructure Inspection Personnel – Bridge** Competency Certificate No. PRP.260897.73040
- **High-Speed Railway Infrastructure Inspection Personnel – Tunnel** Competency Certificate No. PRP.260897.73041

Ministry of Manpower, Republic of Indonesia

- Fire Emergency Response Team (Class C) Registration No. 2794020224 | Valid until 2029
- Fire Emergency Officer (Class D) Registration No. 2813020224 | Valid until 2029

Professional Development

- Railway Infrastructure Inspector Refresher Training
- Improvement of Railway Bridge Management System and Operation Facility
- Lesson Learned: Tunnel Construction in Indonesia

EDUCATION

Politeknik Perkeretaapian Indonesia (PPI) Madiun
Diploma (D3) – Railway Civil Engineering
Major: Railway Track & Civil Infrastructure Engineering
2016 – 2019
GPA: 3.35 / 4.00

SOFTWARE PROFICIENCY

- Microsoft Office (Word, Excel, PowerPoint)
- AutoCAD (Basic)
- PDF Documentation & Technical Reporting

LANGUAGES

- **Bahasa Indonesia** — Native Proficiency
- **English** — Professional Working Proficiency